

2024 AMC 10A Problem 3 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 3. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 3

Problem:

Let n be the least prime number that can be written as the sum of 5 distinct prime numbers. What is the sum of the digits of n ?

Answer Choices:

- A. 5
- B. 7
- C. 8
- D. 10
- E. 11

Solution:

The prime 2 cannot be among the 5 distinct primes chosen because, if it were, then the sum would be even. The first 5 odd primes are 3, 5, 7, 11, and 13, and their sum is 39, which is not prime. The next smallest sum of 5 distinct odd primes is $3 + 5 + 7 + 11 + 17 = 43$, which is prime. The requested digit sum is $4 + 3 = 7$. (B) 7.

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